

DANMARKS TEKNISKE UNIVERSITET



Project Report

02445 - PROJECT IN STATISTICAL EVALUATION OF ARTIFICIAL
INTELLIGENCE AND DATA

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1 Abstract

As large language models (LLMs) are increasingly integrated into educational and communication tools, their ability to adapt explanations to different age groups becomes critical. This project evaluates the ability of 3 different LLMs: ChatGPT 4o, Claude Sonnet 3.7, and Gemini 2.0 Flash , in generating age-appropriate explanations by comparing their fluency on explaining the same topics to distinct age groups of 10, 25 and 50 years old.

To conduct this evaluation, a dataset was constructed by prompting each model with a standardized query format and collecting responses across 64 diverse concepts ranging from scientific topics to political themes. Both qualitative and quantitative methods were employed to evaluate fluency. Human raters assessed the vocabulary, tone, analogy and clarity using a 5-point Likert scale. While automated metrics such as Flesch Reading Ease, Flesch-Kincaid Grade Level, and perplexity scores computed with Hugging Face GPT-2 model provided objective insights into readability and linguistic naturalness. Following that, statistical evaluations such as the two-way ANOVA test were performed to identify significant patterns across models and age groups.

The findings from our evaluation suggest that while large language models (LLMs) are capable of generating generally fluent explanations across age groups, no statistically significant differences in fluency were observed between models or age demographics. This indicates that current LLMs may already achieve a baseline level of age-adaptability in their language, though further refinements are needed to enhance age-specific tailoring. This study contributes to the broader efforts to assess and improve the age adaptability and inclusiveness of AI-generated content.

2 Introduction

Large language models (LLMs), like ChatGPT, Claude and Gemini are now used widely on digital platforms such as classrooms, messaging apps and digital assistants. Their core role is often to break down complex topics for users with vastly different backgrounds and language skills. Given this diversity, the models must produce responses that are not only accurate but also clear and easy to understand. When there is a mismatch between a user's learning needs and the generated response, this can hinder understanding, lead to frustration and ultimately cause users to lose trust in the model or abandon it altogether.

A critical but often overlooked aspect of LLMs is their ability to adapt explanations for different age groups. Children, working adults and older individuals process information differently, respond to distinct vocabulary levels and prefer varying tones. A language model that adjusts its explanations by applying appropriate vocabulary and sentence length while retaining the core information can enhance clarity and utility for all. For instance, simplifying language for a child or using more familiar expressions for an older adult can significantly improve comprehension and engagement. However, it remains unclear whether current models can achieve this level of age-sensitive adaptation or if they favor one age group over the rest.

Hence, this project aims to examine how well different leading language models adjust the fluency of their explanations to suit users of different ages and how well the models compare to each other.

The goals of the project are:

- To collect a diverse set of model outputs across models, topics, and age groups.
- To evaluate the fluency of these responses using a combination of human ratings and quantitative metrics.

- To compare the degree to which different LLMs adapt their language fluency to different age groups.
- To use statistical analysis to determine whether differences in fluency are statistically significant across age groups and models.

3 Methods

This section outlines the methodological approach used to evaluate the fluency of LLMs when generating age-adapted explanations. It covers how data was collected using prompts across multiple age groups and models, how responses were evaluated using human ratings, and how the resulting data were analyzed through descriptive and inferential statistics.

3.1 Data Collection

To evaluate fluency in an age-sensitive context, we created a structured dataset of LLM-generated explanations by prompting models to explain various concepts to individuals of specific ages. We designed our prompts using a simple yet targeted template:

"I am {age} years old. Can you explain {concept} to me?"

The prompt was phrased in first person to create the impression that the model is directly addressing an individual of the stated age, thereby encouraging age-appropriate language and tone.

We then picked three ages; **10, 25, and 50 years old**, representing childhood, full adulthood, and middle age respectively. These age groups were chosen because individuals at different stages of life exhibit varying levels of cognitive development and comprehend information in distinct ways. A ten-year-old looks for simple words and brief sentences, while a fifty-year-old may look for more polished and nuanced writing.

To ensure that the LLMs were evaluated on a broad range of concepts, we used ChatGPT to generate a list of 64 diverse explanation topics. This number was chosen based on a sample size estimation derived from a pilot experiment, which indicated that approximately 190 samples per model were needed to achieve sufficient statistical power. Since the evaluation involves three different age groups, 64 unique topics would yield 192 prompts per model (64 topics $\times$ 3 age groups). The prompt used to generate these topics requested for a mix of subjects spanning categories such as science, technology, and politics (refer to Appendix C). Both concrete concepts (e.g. World War II) and abstract topics (e.g. Free Will) were included as well. The topics generated were returned as a CSV file, allowing for efficient integration into automatic prompt generation. The full list can be seen in the table in Appendix B.

For each prompt, we queried three LLMs:

- **ChatGPT 4o**
- **Claude Sonnet 3.7**
- **Gemini 2.0 flash**

This was done by using the official API's for each model: ChatGPT (OpenAI) [4], Claude (Anthropic) [1], and Gemini (Google) [2]. This made it easy to generate responses automatically, control parameters like age and topic, and avoid any influence from previous interactions. Unlike regular chat interfaces, API calls are stateless and do not carry over context. This ensures consistent behavior across all prompts.

For all models, we used the same setup: a temperature of 0.7 and a batch size of 10. Each

response was saved along with its corresponding model, topic, age group, and the full prompt. The final dataset was stored in a CSV file for later evaluation.

3.2 Evaluation Process

Following data collection, we will use qualitative and quantitative metrics to ensure a comprehensive evaluation.

- **Qualitative metrics:**

- Two independent human raters evaluated each explanation based on 4 fluency dimensions of Vocabulary, Tone, Analogy and Clarity using a 5-point Likert scale. During the evaluation process, each evaluator was presented with the prompt and the corresponding model-generated response. However, the identity of the model (i.e., whether it was ChatGPT, Claude, or Gemini) was deliberately hidden to mitigate potential bias. Additionally, the order in which the samples were shown was randomized for each evaluator.

This setup provided us with a consistent dataset of model-generated explanations evaluated by human raters. The scores for each fluency dimension are then aggregated to compute an overall fluency score for each response. Each fluency dimension and point is defined as follows:

Dimension	Description
Vocabulary	Appropriateness of words and expressions for the target age group.
Tone	Friendliness, formality, and suitability of tone for the intended age.
Analogy	Use of relatable, age-appropriate comparisons or examples, if applicable.
Clarity	How clear and logically structured the explanation is for the given age.

Table 1: Fluency dimensions used for scoring model responses.

Score	Description	Interpretation for Age-Appropriate Fluency
1	Very Poor Fluency	Extremely unnatural, difficult to read, or highly inappropriate for the target age group.
2	Poor Fluency	Noticeably unnatural, somewhat difficult to read, or largely inappropriate for the target age group.
3	Moderate Fluency	Generally readable but with some awkwardness, could be improved to better suit the target age group.
4	Good Fluency	Natural, smooth and easy to read. Mostly appropriate and well-adapted for the target age group.
5	Excellent Fluency	Perfectly natural, highly engaging, and effortless to read. Clear and perfectly adapted to the target age group.

Table 2: 5-point Likert scale

- **Quantitative metrics:**

- Readability scores: The Flesch Reading Ease and Flesch-Kincaid Grade Level scores was computed for each response using the Python library `textstat`. The Flesch

Reading Ease indicates text difficulty, whereby a higher score signifies easier readability. While the Flesch-Kincaid Grade Level provides a U.S. grade-level equivalent of the text's complexity to measure its suitability for specific age groups.

- Perplexity scores: Perplexity measures the fluency and naturalness of the generated explanations. This metric quantifies how well a model predicts a given sequence of words. A lower perplexity score indicates higher predictability and, consequently, greater fluency and coherence of the model. It was computed for all responses using HuggingFace GPT-2 model [3].

3.3 Statistical Methods

After computing relevant metrics, we conducted data analysis and statistical evaluation to identify significant differences in fluency across age-targeted explanations and LLMs.

To prepare the data for statistical analysis, we combined the four fluency dimensions of vocabulary, tone, analogy and clarity by averaging them to create a single overall fluency score for each response. Since each response was rated by two independent evaluators, we also averaged their ratings to obtain one final fluency score per prompt. This averaging helped reduce individual rating bias and ensured a more reliable measurement.

We began by summarizing the data using descriptive statistics, including means and standard deviations of the fluency scores across models and age groups. Boxplots were also created to visualize the distribution of scores and identify any noticeable trends before formal testing.

To test whether fluency differed significantly between models, age groups, or their interaction, we conducted a two-way analysis of variance (ANOVA). This allowed us to examine:

- Whether there are overall differences in fluency between the three models,
- Whether responses differ in fluency depending on the target age group,
- And whether there is a significant interaction effect between model and age group, indicating that some models may adapt better to certain age groups than others.

In addition to the p -values from the ANOVA, we also conducted Kruskal–Wallis tests as a non-parametric alternative to assess differences in fluency scores across models and age groups. This helped validate the robustness of our findings without relying on the assumption of normally distributed residuals.

4 Results

This section presents the results of our evaluation of LLMs on their ability to generate fluent and age-appropriate explanations. We begin by briefly summarizing the pilot study that informed our sampling strategy. Next, we report descriptive statistics to provide an overview of the collected ratings. This is followed by inferential analyses using two-way ANOVA and Kruskal–Wallis. Additional metrics such as readability and perplexity are also examined to complement subjective evaluations.

Pilot Experiment

A small-scale pilot experiment was conducted to test the feasibility of evaluating LLM-generated explanations across age groups and to inform the design of the main study. Using a limited set of prompts and three models (ChatGPT, Claude, and Microsoft Copilot),

we observed generally high fluency scores across age groups, but with slight signs of lower performance for the oldest age group. Based on these findings, we revised the setup for the final experiment: we replaced Copilot with Gemini, as Copilot and ChatGPT seemed to provide very similar responses, and adjusted the age groups from 8, 25, and 70 to 10, 25, and 50 to better reflect typical users. The pilot results also served as a basis for our sample size estimation. The results of the pilot experiment can be seen in appendix D.

4.1 Descriptive Statistics

To provide an overview of how fluency ratings varied across age groups and language models, we computed the mean and standard deviation of fluency scores for each model-age combination. Table 3 shows the results in a structured format, while figure 1 displays the distributions of fluency ratings using a boxplots. The mean and standard deviation of fluency scores by model and by age group, as well as the corresponding boxplots, can be found in appendix E.

Model	Age 10	Age 25	Age 50
Claude 3.7 Sonnet	4.67 (0.22)	4.63 (0.24)	4.58 (0.24)
GPT-4o	4.63 (0.23)	4.67 (0.19)	4.65 (0.25)
Gemini 2.0 Flash	4.70 (0.22)	4.64 (0.24)	4.62 (0.25)

Table 3: Mean fluency score, with standard deviation in parenthesis, by model and age group

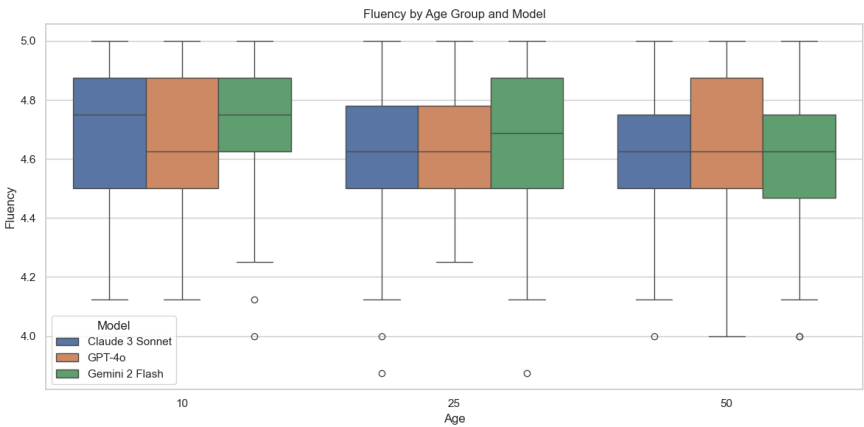


Figure 1: Boxplot of fluency ratings by age group and model

Overall, the differences in fluency ratings across models and age groups are relatively minor, and all models appear to perform well in terms of fluency, with average scores consistently above 4.5. Nonetheless, some small patterns can be observed. Claude exhibits a slight downward trend across age groups, achieving its highest average fluency at age 10 (4.67) and gradually decreasing to 4.58 by age 50. A similar trend is visible for Gemini, which also scores highest for the youngest group and slightly lower for older participants. In contrast, chatGPT demonstrates the most stable performance across ages, with mean fluency scores remaining nearly constant.

The boxplot in figure 1 supports this observation, showing tightly grouped fluency scores for all model-age combinations with limited variability.

4.2 Readability and Perplexity Scores

To complement the fluency-based evaluation, we analyzed the readability and language model perplexity of the LLM-generated responses. Specifically, we computed the *Flesch Reading Ease* (FRE) and *Flesch-Kincaid Grade Level* (FKG) as standard measures of linguistic accessibility, and perplexity as an estimate of language model fluency and confidence. The results are presented in appendix F in tables 9, 10, and 11.

Across all three metrics, ChatGPT consistently delivers responses that are both easier to read (highest FRE, lowest FKG) and more predictable (lowest perplexity), especially for younger audiences. Claude shows strong readability at age 10, but the complexity increases dramatically with age, with FRE dropping and FKG rising sharply for age 25 and 50. Gemini generally performs between the two, with modest readability and perplexity scores.

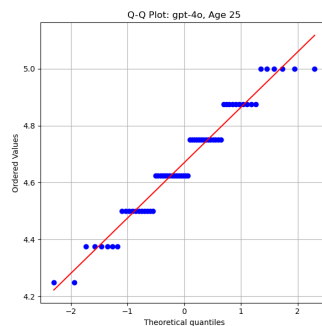
These findings suggest that while all models adapt output to different ages, only GPT-4o achieves consistently high readability and fluency, especially in tailoring responses to younger users.

4.3 Inferential Statistics

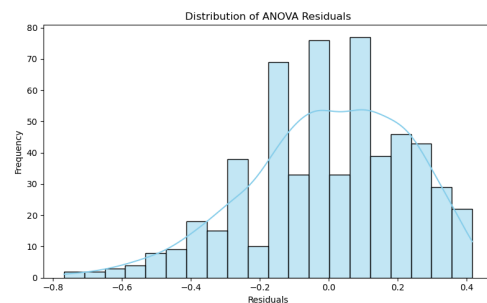
To assess whether the differences in fluency ratings across language models and age groups were statistically significant, we conducted a two-way ANOVA. Prior to this, we evaluated whether the assumptions underlying ANOVA were sufficiently met.

Normality of residuals. To check the normality assumption, we inspected Q-Q plots of fluency scores within each group and the distribution of residuals from the ANOVA model. All Q-Q plots followed the same tendency and seemed to suggest approximate normality with some deviations. Figure 2a shows one representative example (GPT-4o, Age 15).

In addition, the overall distribution of residuals can be seen in figure 2b). The distribution of residuals appeared to maintain a roughly bell-shaped form, though noticeable deviations from perfect normality were present. Specifically, the distribution showed irregularities that suggest it is not strictly normal. Nevertheless, due to the sufficiently large sample size and the fact that ANOVA is generally robust to moderate violations of normality, we proceeded with the two-way ANOVA analysis.



(a) Q-Q plot for GPT-4o at Age 25.



(b) Distribution of residuals from the two-way ANOVA model.

Figure 2: Visual checks for the normality assumption: a Q-Q plot (left) and residual distribution (right).

Two-way ANOVA results. The two-way ANOVA revealed no statistically significant effects. As shown in Table 4, there were no significant differences in fluency scores between models ($p = 0.51$) or between age groups ($p = 0.13$). Furthermore, the interaction between model and age group also failed to reach significance ($p = 0.32$), indicating that the effect of model did not vary meaningfully across different age groups. Overall, these results suggest that neither model choice, target age group, nor their interaction had a statistically reliable influence on fluency scores.

Table 4: Two-way ANOVA results for fluency scores

Effect	Sum Sq	df	F	p-value
Model	0.073	2	0.68	0.51
Age	0.216	2	2.02	0.13
Model $\times$ Age	0.254	4	1.19	0.32

Kruskal-Wallis. Due to observed deviations from normality in the residuals and Q-Q plots, we also performed Kruskal-Wallis tests to validate the robustness of the ANOVA results. This test does not assume normality and is suitable for ordinal or non-normally distributed interval data like our averaged fluency scores.

The results of Kruskal-Wallis tests comparing fluency scores across the three models for each age group is shown in table 12 in appendix G. None of the comparisons yielded statistically significant results ($p > 0.05$), supporting the earlier ANOVA findings. The p-values ranged from 0.23 to 0.92, indicating no strong evidence for differences in model fluency performance within any particular age group.

The Kruskal-Wallis test results comparing fluency scores across different age groups for each model can be seen in table 13 in appendix G. Again, none of the models showed statistically significant variation across age groups. Claude-3.7 Sonnet and Gemini 2.0 Flash showed slight trends ($p = 0.12$ and $p = 0.19$, respectively), but these effects did not reach significance. This further reinforces the conclusion that fluency scores were relatively stable across both models and age groups.

Overall, these results align closely with the findings from the two-way ANOVA, indicating that neither model nor age group significantly influenced the fluency ratings, and there was no interaction effect.

5 Discussion

This project illustrated that LLMs such as ChatGPT 4o, Claude Sonnet 3.7, and Gemini 2.0 Flash have similar capabilities in generating fluent and age-adapted explanations across a wide range of topics.

By combining subjective human ratings on vocabulary, tone, analogy, and clarity, along with objective automated metrics like readability and perplexity scores, a thorough evaluation framework was formed. This mixed-methods strategy effectively captured both the subtle elements of language fluency and quantifiable linguistic features. The pilot experiment confirmed that this approach was feasible and validated that the chosen fluency dimensions were significant and could be assessed reliably.

Upon computing descriptive statistics and two-way ANOVAs, these analysis revealed no statistically significant main effects for either the model or age group, nor any significant

interaction effects. This suggests that none of the evaluated models consistently outperformed the others in terms of fluency, regardless of the target age. The models also did not appear to adjust the fluency of their responses significantly different manner based on whether they were addressing a child, an adult, or a middle-aged user.

However, the absence of statistically significant difference does not necessarily imply that no real differences exist. Rather, it may indicate that any existing differences were too subtle to be detected by our sample size or evaluation method. In this case, our dataset and evaluation design may lack the sensitivity to capture more nuanced distinctions. Future work and improvements can be done to address these limitations, ensuring a more detailed and comprehensive evaluation (refer to Section 5.1).

5.1 Limitations and Future Work

The study included two human raters, whose evaluations offered valuable initial insights. However, the small number and limited age range of the evaluators constrain the generalizability of subjective fluency ratings. Expanding the evaluator group to include a larger and more varied demographic would enhance reliability, reduce personal bias and offer a more representative assessment of how fluency is perceived by various age groups.

Furthermore, the evaluation relied on direct, standardized prompts in a one-turn format, which may not capture the contextual aspects of real-life scenarios. Future studies could explore how age adaptation functions within interactive dialogues and more naturalistic settings to provide a deeper understanding of model performance.

6 Conclusion

In conclusion, the study examined the ability of 3 leading LLMs (ChatGPT 4o, Claude Sonnet 3.7, and Gemini 2.0 Flash) in adapting the fluency of their explanations for different age groups. Across 64 topics and 3 age profiles, we assessed their outputs using human ratings, automated metrics and statistical tests.

All three models demonstrated high fluency and general capability in tailoring explanations to different ages. ChatGPT-4o showed the most consistent performance especially in producing readable content for younger audiences. While Claude displayed a progression in complexity with increasing age. However, statistical analysis found no significant differences in fluency between models or age groups.

These findings highlight that current LLMs can generate age-appropriate content with strong baseline quality. Yet, there is room for growth especially in better differentiating content for older age groups and refining sensitivity to audience needs. As LLMs play a growing role in education and communication tools, improving their ability to tailor content to specific audiences will be essential for ensuring their effectiveness and impact.

References

- [1] Anthropic. Claude api. <https://docs.anthropic.com/en/home>, 2025. Accessed: 2025-06-18.
- [2] Google. Gemini api. <https://ai.google.dev/gemini-api/docs>, 2025. Accessed: 2025-06-18.
- [3] HuggingFace. Perplexity of fixed-length models. <https://huggingface.co/docs/transformers/perplexity>, 2025. Accessed: 2025-06-18.
- [4] OpenAI. Chatgpt api. <https://platform.openai.com/docs/overview>, 2025. Accessed: 2025-06-17.

A Link to GitHub repository

https://github.com/RallerR/02445_project

B List of topics

Gravity	Photosynthesis	DNA	The water cycle
Electricity	States of matter	Atoms	Magnetism
Evolution	Sound waves	Artificial Intelligence	The Internet
How computers work	Cybersecurity	Social media	Robots
Machine learning	Smartphones	Virtual reality	Democracy
Taxes	Human rights	Immigration	Freedom of speech
Justice	Elections	Police systems	Governments
Social equality	Vaccines	Nutrition	Exercise
Mental health	Sleep	Diseases	Medicines
Hygiene	The human body	World War II	The Cold War
Ancient Egypt	The Roman Empire	Slavery	Industrial Revolution
Civil Rights Movement	The Vikings	Colonialism	Climate change
Recycling	Pollution	Renewable energy	Deforestation
Biodiversity	Carbon footprint	Sustainability	Water scarcity
Supply and demand	Inflation	Capitalism	Ethics
Utilitarianism	Free will	Markets	Scarcity

Table 5: List of 64 Topics Used for Prompt Generation

C Prompt for generating list of topics

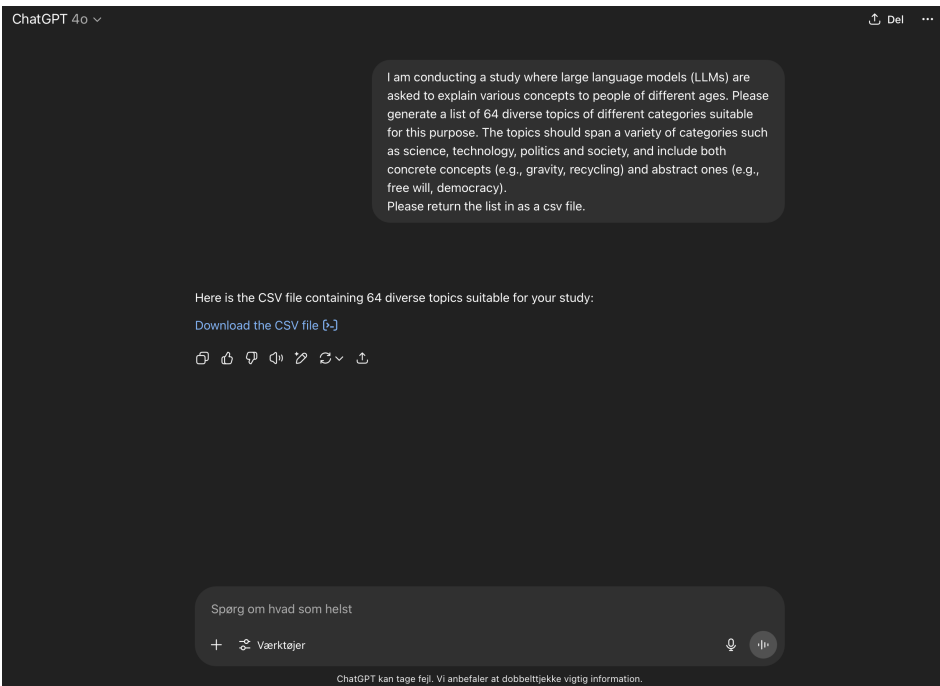


Figure 3: Prompt used in ChatGPT to generate 64 different concepts/topics.

D Results of pilot experiment

An overview of the fluency scores for each model and age group is shown in the table 6.

Age	ChatGPT 4o	Claude Sonnet 4	Copilot
8	4.58	4.50	4.64
30	4.56	4.72	4.67
70	4.50	4.47	4.58

Table 6: Fluency scores per model across different age groups.

E Descriptive Statistics for models and age

Model	Fluency (Mean $\pm$ SD)
Claude 3.7 Sonnet	4.63 (0.23)
GPT-4o	4.65 (0.22)
Gemini 2.0 Flash	4.65 (0.24)

Table 7: Mean fluency scores with standard deviation for each model

Age Group	Fluency (Mean $\pm$ SD)
10	4.67 (0.22)
25	4.65 (0.23)
50	4.62 (0.24)

Table 8: Mean fluency scores with standard deviation for each age group

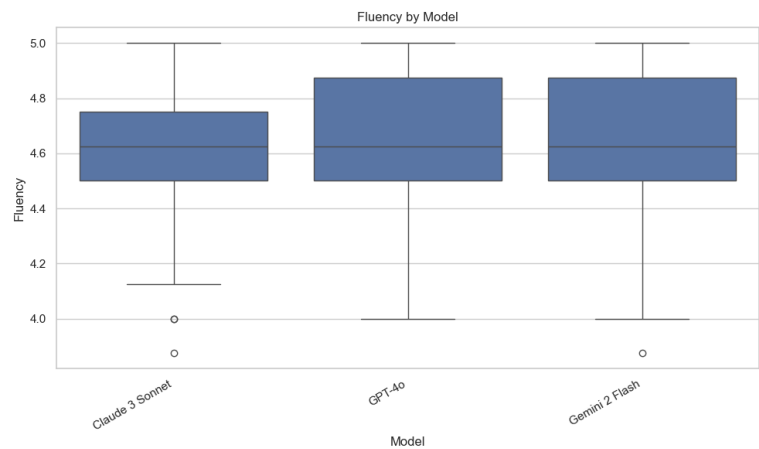


Figure 4: Boxplot of fluency ratings by model

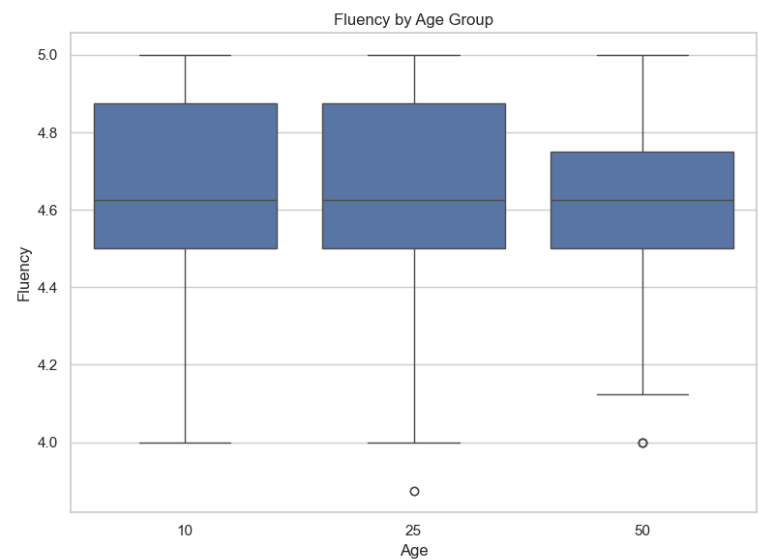


Figure 5: Boxplot of fluency ratings by age group

F Readability and Perplexity Scores

Model	Age 10	Age 25	Age 50
Claude 3 Sonnet	57.42 (13.07)	2.64 (24.94)	6.73 (28.75)
GPT-4o	68.20 (7.83)	32.28 (12.09)	37.62 (13.56)
Gemini 2 Flash	64.42 (8.36)	30.61 (9.85)	32.86 (10.76)

Table 9: Mean and standard deviation of Flesch Reading Ease for each model and age.

Model	Age 10	Age 25	Age 50
Claude 3 Sonnet	11.31 (4.30)	23.32 (8.27)	22.05 (8.78)
GPT-4o	7.79 (1.41)	12.98 (1.93)	12.12 (1.92)
Gemini 2 Flash	8.38 (1.35)	14.38 (2.19)	13.97 (2.40)

Table 10: Mean and stardard deviation of Flesch-Kincaid Grade Level for each model and age.

Model	Age 10	Age 25	Age 50
Claude 3 Sonnet	20.83 (5.37)	21.32 (6.16)	20.84 (5.23)
GPT-4o	11.13 (2.17)	10.50 (2.01)	10.50 (2.04)
Gemini 2 Flash	14.17 (3.36)	13.56 (3.29)	13.57 (3.49)

Table 11: Mean and standard deviation of perplexity scores for each model and age.

G Kruskal-Wallis test

Table 12: Kruskal-Wallis test comparing models at each age group

Age	H-statistic	p-value
10	2.9653	0.2270
25	0.1672	0.9198
50	2.6473	0.2662

Table 13: Kruskal-Wallis test comparing age groups within each model

Model	H-statistic	p-value
Claude 3.7 Sonnet	4.1823	0.1235
GPT-4o	0.7419	0.6901
Gemini 2.0 Flash	3.2761	0.1944